

Experiment 1a: 6-Action Extension

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1 Motivation

The MVP simulator uses a binary action space (`nudge`, `idle`). Issue #126 extends this to 6 clinically-motivated actions with per-action metadata (`reward_penalty`, `burden_penalty`) while maintaining full backward compatibility with `list[str]` configs.

2 Config Schema

Actions now accept two formats:

- `list[str]`: MVP mode, e.g. `[nudge, idle]`
- `list[dict]`: Extended mode with metadata:
 - `name`: `motivational_prompt`
 - `reward_penalty`: `0.1`
 - `burden_penalty`: `0.2`

The `_normalize_actions()` validator extracts names, rejects mixed types and duplicates. Agents receive the flat `action_names` list.

3 6-Action Config

Action	Reward Penalty	Burden Penalty
<code>no_message</code>	0.00	0.00
<code>motivational_prompt</code>	0.10	0.20
<code>walking_suggestion</code>	0.20	0.30
<code>goal_reminder</code>	0.05	0.15
<code>recovery_suggestion</code>	0.15	0.25
<code>progress_feedback</code>	0.08	0.10

4 Results (50 seeds, 90 days)

Agent	Total Reward	Per Step
Standard TS	260.8 ± 23.8	0.5796
Contextual TS	248.8 ± 21.2	0.5529
Standard EG	260.1 ± 33.0	0.5781
Contextual EG	249.4 ± 30.8	0.5543
Standard UCB	270.4 ± 24.5	0.6009
Contextual UCB	254.4 ± 19.6	0.5653
Standard DEC	268.6 ± 32.5	0.5969
Contextual DEC	255.7 ± 25.0	0.5681
Random	211.4 ± 17.1	0.4699

Standard UCB achieves the highest reward (270.4). Contextual agents do not outperform standard variants on this config, likely because the 6-action space is too large for activity-only conditioning to learn effectively within 450 steps.

5 Open Questions

- The penalty values are hand-tuned placeholders. A grid search over penalty combinations could identify regions where burden penalty meaningfully shapes agent behaviour.
- With 6 actions and only 2 states, the transition matrix has 12 entries. Real clinical data would inform more realistic transition probabilities.